Algorithm

An **algorithm** is a well-defined procedure that allows a computer to solve a problem.

- a sequence of unambiguous instructions.

Algorithms design

Logical Thinking: Methods of reasoning, a set of rules and techniques to determine whether a statement is valid.

- chip design
- safety-critical software
- network-secure software
- all software development
- artificial intelligence
- database systems, etc.

Algorithms design

Logical Thinking: Methods of reasoning, a set of rules and techniques to determine whether a statement is valid.

- **Propositional logic**: Rules of logic that give precise meaning to mathematical statements.

- **Predicate logic**: Rules of logic that express the meaning of a wide range of statements in mathematics and computer science in ways that permit us to reason and explore relationships between objects.

- **Proof Techniques**: Various proof techniques for claiming a fact.

Algorithms design

Relational Thinking

- **Sets**: Collections of well-defined objects.
- **Relations**: Relations among objects.
- **Functions**: Illustration of various mappings.

Algorithms design

Recursive Thinking: Get a solution for the large problem with the solutions of smaller sub-(sub)problems.

- Recursion (Climbing stairs: Climb one stair + solve the rest)
- Induction (Prove: $2^n \geq n+1 \ \forall n \geq 0$)
- Reduction (Sum(n) = n + Sum(n-1))

Algorithms design

Quantitative Thinking: It uses numbers to make better decisions in real life.

- Counting Principles
- Arithmetic (addition, subtraction,…)
- Planning, performance, average, median, estimation,…

Algorithms design

Analytical Thinking

- Functions and Recursive Algorithms
- Iterative Algorithms
- Optimization, Quantification, Analysis

Proof – establishment the truth of a mathematical statement

- In mathematics, a *proof* is:
  - a *correct* (well-reasoned, logically valid) and *complete* (clear, detailed) argument that rigorously and undeniably establishes the truth of a mathematical statement.

- Why must the argument be correct and complete?
  - *Correctness* prevents us from fooling ourselves.
  - *Completeness* allows anyone to verify the result.